Problem Set 7 - Dictionaries

Hello students, the goal of this problem-set is to familiarize yourself with the *dictionaries*.

You can find all the methods associated with dictionaries in the slides and on the [Official Documentation].

Assignment 7.1 - Black Friday Wishlist (optional)

Write a program, that uses a dictionary as a storage structure, to implement a wish list.

The program must continuously present the user with a menu to allow the user to perform different operations.

The first operation allowed is the insertion of a new item in the wishlist (product name and price).
If an item with the same name is already present in the wishlist, the program must avoid adding it again but should print a warning and go on.

The second option available must allow the user to print the list of available items.

The third option should request the maximum budget to the user, and then print the list of affordable items.

The last option allows the user to quit the program.

Here an example of the application run. The output generated must look like this (inputs are marked in **bold red**):

```
1. Add an item to the wishlist
2. List items in the wishlist
3. Insert your budget and print the affordable items
4. Quit

What operation you want to perform? 1
Insert the product name: Monitor
Insert the product price: 199.99
New product added to the wishlist!

1. Add an item to the wishlist
2. List items in the wishlist
3. Insert your budget and print the affordable items
4. Quit

What operation you want to perform? 1
Insert the product name: Keyboard
Insert the product price: 99.90
New product added to the wishlist!
```

```
1. Add an item to the wishlist
2. List items in the wishlist
3. Insert your budget and print the affordable items
4. Quit

What operation you want to perform? 1
Insert the product name: Monitor
Insert the product price: 199.99
Product already added!

1. Add an item to the wishlist
2. List items in the wishlist
3. Insert your budget and print the affordable items
4. Quit

What operation you want to perform? 2
List of all the items in store:
Monitor : 199.99
Keyboard : 99.9

1. Add an item to the wishlist
2. List items in the wishlist
3. Insert your budget and print the affordable items
4. Quit

What operation you want to perform? 3
Enter the maximum price: 120

List of affordable items:
Keyboard : 99.9

1. Add an item to the wishlist
2. List items in the wishlist
3. Insert your budget and print the affordable items
4. Quit

What operation you want to perform? 4
Bye bye!
```

Optional: Add a new voice in the menu that allows you to choose an element to be deleted from the wishlist

Assignment 7.2 - Translator (optional)

Write a program that translates words from one language to another (e.g., English to Italian).

The program will ask the user to insert a list of **five words** in one language and then the same words in the destination language. The words must be saved into a dictionary.

The program will then allow the user to insert any word in the first language and will print the corresponding translation.

If the word to translate is new, the program should ask for its translation and update its internal dictionary. The program will continue to run until the user inserts `!stop`.

For example (inputs are marked in **bold red**):

```
List of words in the first language:
Word 1: Cat
Word 2: Student
Word 3: History
Word 4: Twenty
Word 5: Information

Equivalent in the second language:
(Cat): Gatto
(Student): Studente
(History): Storia
(Twenty): Venti
(Information): Informazione

Word to translate: Information
Translated word: Informazione

Word to translate: Ham
"Ham" is a new word, insert the translation: Prosciutto
Dictionary updated: Ham -> Prosciutto!

Word to translate: Ham
Translated word: Prosciutto

Word to translate: !stop
Stopped
```

Assignment 7.3 - Characters counter (optional)

Write a program that asks the user for a sentence. The program must then count how many times each character appears, and print the result in alphabetical order.

The output should look like this (inputs are marked in **bold red**):

```
Text: Master Yoda...  is Darth Vader my father?  Mmm...  rest I need.

Word counter:
  : 10
. : 7
? : 1
a : 5
d : 4
e : 6
f : 1
h : 2
i : 2
m : 5
n : 1
o : 1
r : 5
s : 3
t : 4
v : 1
y : 2
```

Assignment 7.4 - Factorial (optional)

Write a program that asks the user for a number and calculates the factorial. The program stops the execution when the user inserts a number lower than 0.
The program must use dictionaries to improve its performance every time the user inserts a new number by caching (saving) the previous results.

The output should look like (inputs are marked in **bold red**):

```
Calculate factorial: 5
Result of 5! calculated in 0.03337860107421875 ms!
Calculate factorial: 5000
Result of 5000! calculated in 29.19292449951172 ms!
Calculate factorial: 5000
Result of 5000! calculated in 0.019788742065429688 ms!
Calculate factorial: 7000
Result of 7000! calculated in 8.979320526123047 ms!
Calculate factorial: 7012
Result of 7012! calculated in 0.1747608184814453 ms!
Calculate factorial: -1
```

Note: to print the time execution insert *import time* before the main. Then you save the start time using *start_time = time.time()* and calculate the difference between the end and the start in seconds using *time.time() - start_time*